

Math Thermo
class #10
02/19/2026

last time we saw from an example that

where
$$\beta = \frac{1}{T} = \frac{\partial S_{eq}}{\partial U},$$

$$S_{eq}(U, N) := S_G(\vec{V}^{eq}).$$

let us try another simple example.

Example (10.1): let us look at another simple two-state system. Suppose that

$$U_1 = \varepsilon, \quad U_2 = 2\varepsilon.$$

The constraints are

$$n_1 + n_2 = N$$

$$n_1 \varepsilon + n_2 2\varepsilon = U$$

Again, we can solve for n_1 and n_2 in terms of the parameters since there are exactly two constraint equations.

$$n_1 = N - n_2$$

$$\varepsilon(N - n_2) + n_2 \cdot 2\varepsilon = U$$

$$\Rightarrow n_2 \varepsilon = U - \varepsilon N$$

$$\Rightarrow n_2 = \frac{U - \varepsilon N}{\varepsilon} = \frac{U}{\varepsilon} - N$$

Thus,

$$n_1^{\text{eq}} = 2N - \frac{U}{\varepsilon}, \quad n_2^{\text{eq}} = \frac{U}{\varepsilon} - N.$$

We can calculate the Gibbs entropy at equilibrium directly

$$\begin{aligned} S_{\text{eq}}(U, N) &= S_G(\vec{n}_{\text{eq}}) \quad (\vec{n}_{\text{eq}} = (n_1^{\text{eq}}, n_2^{\text{eq}})) \\ &= -k_B \sum_{i=1}^2 n_i^{\text{eq}} \log\left(\frac{n_i^{\text{eq}}}{N}\right) \\ &= -k_B \left(\frac{2N\varepsilon - U}{\varepsilon}\right) \log\left(\frac{2N\varepsilon - U}{N\varepsilon}\right) \\ &\quad - k_B \left(\frac{U - N\varepsilon}{\varepsilon}\right) \log\left(\frac{U - N\varepsilon}{N\varepsilon}\right). \end{aligned}$$

As above, let us compute the temperature

$$\begin{aligned} \frac{1}{T} &= \frac{\partial S_{\text{eq}}}{\partial U} \\ &= \frac{k_B}{\varepsilon} \log\left(\frac{2N\varepsilon - U}{N\varepsilon}\right) \\ &\quad - k_B \left(\frac{2N\varepsilon - U}{\varepsilon}\right) \frac{N\varepsilon}{2N\varepsilon - U} \left(-\frac{1}{N\varepsilon}\right) \end{aligned}$$

$$\begin{aligned}
 & -\frac{k_B}{\epsilon} \log \left(\frac{U - N\epsilon}{N\epsilon} \right) \\
 & -k_B \left(\frac{U - N\epsilon}{\epsilon} \right) \frac{N\epsilon}{U - N\epsilon} \frac{1}{N\epsilon} \\
 & = \frac{k_B}{\epsilon} \log \left(\frac{2N\epsilon - U}{N\epsilon} \cdot \frac{N\epsilon}{U - N\epsilon} \right).
 \end{aligned}$$

Thus,

$$\frac{1}{T} = \frac{k_B}{\epsilon} \log \left(\frac{2N\epsilon - U}{U - N\epsilon} \right).$$

Now, let us use Theorem (9.9) again. The partition function is

$$Z = \sum_{i=1}^2 \exp \left(-\frac{\beta U_i}{k_B} \right)$$

$$= \exp \left(-\frac{\beta \epsilon}{k_B} \right) + \exp \left(-\frac{2\beta \epsilon}{k_B} \right).$$

The equilibrium n_i are given by

$$\frac{n_1^{eq}}{N} = \frac{\exp \left(-\frac{\beta \epsilon}{k_B} \right)}{\exp \left(-\frac{\beta \epsilon}{k_B} \right) + \exp \left(-\frac{2\beta \epsilon}{k_B} \right)}$$

and

$$\frac{n_2^{eq}}{N} = \frac{\exp \left(-\frac{2\beta \epsilon}{k_B} \right)}{\exp \left(-\frac{\beta \epsilon}{k_B} \right) + \exp \left(-\frac{2\beta \epsilon}{k_B} \right)}$$

Of course, we can plug these into our Gibbs entropy formula. But once again, we are after the value of β !

We find β by using the energy constraint:

$$n_1^e \epsilon + 2n_2^e \epsilon = U.$$

Thus

$$\begin{aligned} N\epsilon \exp\left(-\frac{\beta\epsilon}{k_B}\right) + 2N\epsilon \exp\left(-\frac{2\beta\epsilon}{k_B}\right) \\ = Z \cdot U \\ = U \left[\exp\left(-\frac{\beta\epsilon}{k_B}\right) + \exp\left(-\frac{2\beta\epsilon}{k_B}\right) \right]. \end{aligned}$$

Set

$$\beta' := \frac{\beta\epsilon}{k_B}$$

for simplicity. Then,

$$1 + 2 \exp(-\beta') = \frac{U}{N\epsilon} [1 + \exp(-\beta')]$$

$$\Rightarrow \frac{2N\epsilon - U}{N\epsilon} \exp(-\beta') = \frac{U - N\epsilon}{N\epsilon}$$

$$\Rightarrow \exp(-\beta') = \frac{U - N\epsilon}{2N\epsilon - U}$$

$$\Rightarrow -\beta' = \log \left(\frac{U - N\epsilon}{ZN\epsilon - U} \right)$$

$$\Rightarrow \beta' = \log \left(\frac{ZN\epsilon - U}{U - N\epsilon} \right)$$

$$\Rightarrow \beta = \frac{k_B}{\epsilon} \log \left(\frac{ZN\epsilon - U}{U - N\epsilon} \right) \\ = \frac{1}{T} !$$

Let us prove this in general.

Theorem (10.2): Suppose that the energy levels, U_i , are fixed. At equilibrium, the Gibbs entropy satisfies

$$(10.1) \quad S_{eq}(U, N) := S_G(\vec{v}_{eq}) = \beta U + k_B \log(Z) N, \quad \text{Euler Equation}$$

and at equilibrium the derivatives are

$$(10.2) \quad \frac{\partial S_{eq}}{\partial U} = \beta \quad \text{and} \quad \frac{\partial S_{eq}}{\partial N} = k_B \log(Z).$$

Proof: Recall that, at equilibrium,

$$n_j^{eq} = \exp \left(\frac{-\beta U_j}{k_B} \right) \cdot \frac{N}{Z}$$

where

$$Z = \sum_{i=1}^r \exp \left(\frac{-\beta U_i}{k_B} \right).$$

The Gibbs entropy is, therefore,

$$\begin{aligned} S_{eq}(U, N) &= S_G(\vec{n}_{eq}) \\ &= -k_B \sum_{i=1}^r n_i \log\left(\frac{n_i}{N}\right) \\ &= -\frac{k_B N}{Z} \sum_{i=1}^r \exp\left(\frac{-\beta U_i}{k_B}\right) \log\left(\frac{\exp\left(\frac{-\beta U_i}{k_B}\right)}{Z}\right) \\ &= -\frac{k_B N}{Z} \sum_{i=1}^r \exp\left(\frac{-\beta U_i}{k_B}\right) \left\{ -\frac{\beta U_i}{k_B} - \log(Z) \right\} \\ &= \frac{N\beta}{Z} \sum_{i=1}^r \exp\left(\frac{-\beta U_i}{k_B}\right) U_i \\ &\quad + \frac{k_B N \log(Z)}{Z} \sum_{i=1}^r \exp\left(\frac{-\beta U_i}{k_B}\right) \\ &= \frac{N\beta}{Z} \sum_{i=1}^r \exp\left(\frac{-\beta U_i}{k_B}\right) U_i + k_B N \log(Z) \\ &= \beta \sum_{i=1}^r U_i n_i + k_B N \log(Z) \\ &= \beta U + k_B \log(Z) N. \end{aligned}$$

Now, to find the derivative with respect to energy, suppose that each n_j is a function of the process parameter α :

$$n_j = n_j(\alpha).$$

Then, $N = N(r)$, and since

$$N(r) = \sum_{j=1}^r n_j(r),$$

it follows that

$$N'(r) = \sum_{j=1}^r n_j'(r).$$

From the definition of the Gibbs entropy,

$$\begin{aligned} \frac{d}{dr} S_G(\vec{v}(r)) &= -k_B \frac{d}{dr} \left[\sum_{i=1}^r n_i \log\left(\frac{n_i}{N}\right) \right] \\ &= -k_B \frac{d}{dr} \left[\sum_{i=1}^r n_i (\log(n_i) - \log(N)) \right] \\ &= -k_B \sum_{i=1}^r \left\{ n_i'(r) [\log(n_i(r)) - \log(N(r))] \right. \\ &\quad \left. + n_i(r) \left[\frac{n_i'(r)}{n_i(r)} - \frac{N'(r)}{N(r)} \right] \right\} \\ &= -k_B \sum_{i=1}^r [\log(n_i(r)) - \log(N(r))] n_i'(r) \\ &\quad - k_B \sum_{i=1}^r \left[n_i'(r) - \frac{n_i(r)}{N(r)} N'(r) \right] \\ &= -k_B \sum_{i=1}^r [\log(n_i(r)) - \log(N(r))] n_i'(r) \\ &\quad - k_B N'(r) + k_B N'(r). \end{aligned}$$

Hence,

$$(10.3) \quad \frac{d}{d\vec{r}} S_G(\vec{r}) = -k_B \sum_{i=1}^r \log\left(\frac{n_i(\vec{r})}{N(\vec{r})}\right) n'_i(\vec{r}).$$

This gives information about the gradients, namely,

$$(10.4) \quad \frac{\partial S_G}{\partial n_j} = -k_B \log\left(\frac{n_j}{N}\right),$$

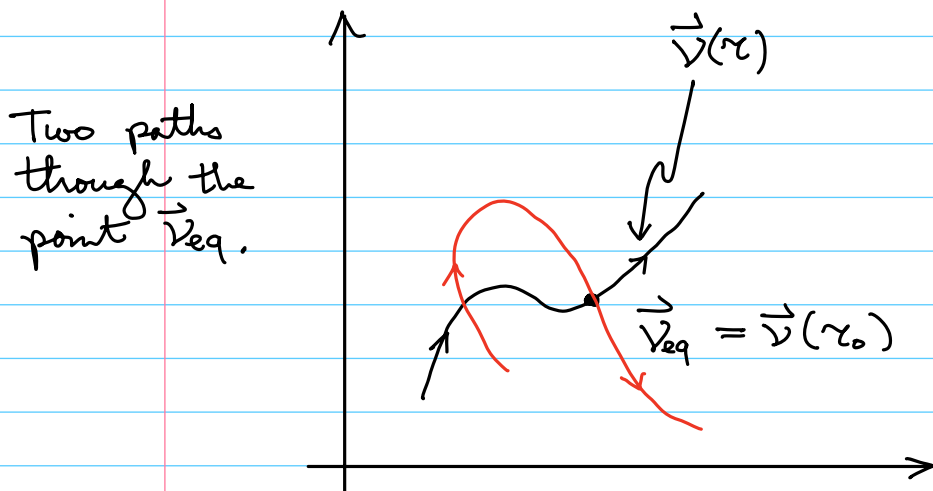
where

$$N = \sum_{i=1}^r n_i.$$

Now, since energy levels are fixed and

$$U(\vec{r}) = \sum_{i=1}^r U_i n_i(\vec{r})$$

$$U'(\vec{r}) = \sum_{i=1}^r U_i n'_i(\vec{r})$$



Now, suppose that the path through state space passes through the equilibrium point $\vec{v}_{eq}$ at $\vec{r} = \vec{r}_0$

Then,

$$\left. \frac{d}{d\vec{r}} S_G(\vec{v}) \right|_{\vec{r}=\vec{r}_0} = -k_B \sum_{i=1}^r \log\left(\frac{n_i(\vec{r}_0)}{N(\vec{r}_0)}\right) n'_i(\vec{r}_0)$$

$$\begin{aligned}
&= -k_B \sum_{i=1}^r \log\left(\frac{n_i^{eq}}{N}\right) n'_i(r_0) \\
&= -k_B \sum_{i=1}^r \log\left(\exp\left(-\frac{\beta U_i}{k_B}\right) \frac{1}{Z}\right) n'_i(r_0) \\
&= -k_B \sum_{i=1}^r \left(-\frac{\beta U_i}{k_B} - \log(Z)\right) n'_i(r_0) \\
&= \beta \sum_{i=1}^r U_i n'_i(r_0) + k_B \log(Z) N'(r_0) \\
&= \beta U'(r_0) + k_B \log(Z) N'(r_0).
\end{aligned}$$

Since the path $\vec{v}(r)$ we choose is arbitrary, apart from the fact that it passes through $\vec{v}_{eq}$, we may consider $U'(r_0)$ and $N'(r_0)$ as arbitrary. Thus,

$$\frac{\partial S_{eq}}{\partial U} = \beta \quad \text{and} \quad \frac{\partial S_{eq}}{\partial N} = k_B \log(Z) \quad . \quad //$$

Definition (10.3): The statistical temperature is defined by the expression

$$(10.5) \quad \frac{1}{T} := \frac{\partial S_{eq}}{\partial U},$$

where

$$S_{eq}(U, N) = S_G(\vec{v}_{eq}).$$

The statistical chemical potential is defined
via

$$(10.6) \quad \frac{\mu}{T} := \frac{\partial S_{eq}}{\partial N}.$$

Thus

$$(10.7) \quad T = \frac{1}{\beta}$$

and

$$(10.8) \quad \mu = k_B T \log(Z).$$

For the equilibrium macrostates we now write

$$(10.9) \quad n_j^{eq} = \exp\left(\frac{-U_j}{k_B T}\right) \cdot \frac{N}{Z}$$

$$(10.10) \quad Z = \sum_{i=1}^r \exp\left(\frac{-U_i}{k_B T}\right).$$

Here, we must keep in mind that, in general,

$$T = T(U, N)$$

and

$$\mu = \mu(U, N).$$

Example (10.4): Recall that, in Example (10.1) we found that, with the 2 energy levels

$$U_1 = \varepsilon \quad \text{and} \quad U_2 = 2\varepsilon,$$

$$S_{eq}(U, N) = -k_B \left(\frac{2N\varepsilon - U}{\varepsilon} \right) \log \left(\frac{2N\varepsilon - U}{N\varepsilon} \right)$$

$$-k_B \left(\frac{U - N\varepsilon}{\varepsilon} \right) \log \left(\frac{U - N\varepsilon}{N\varepsilon} \right)$$

and

$$\frac{1}{T} = \frac{k_B}{\varepsilon} \log \left(\frac{2N\varepsilon - U}{U - N\varepsilon} \right).$$

Observe that S_{eq} is homogeneous of degree 1 and the temperature is homogeneous of degree 0.

Now, given T and N , we may invert the temperature relation to find a unique value of U :

$$\exp \left(\frac{-\varepsilon}{k_B T} \right) = \frac{U - N\varepsilon}{2N\varepsilon - U}$$

$$(2N\varepsilon - U) \exp \left(\frac{-\varepsilon}{k_B T} \right) = U - N\varepsilon$$

$$2N\varepsilon \exp \left(\frac{-\varepsilon}{k_B T} \right) + N\varepsilon = U + U \exp \left(\frac{-\varepsilon}{k_B T} \right)$$

$$U = \frac{\varepsilon N \left(2 \exp \left(\frac{-\varepsilon}{k_B T} \right) + 1 \right)}{1 + \exp \left(\frac{-\varepsilon}{k_B T} \right)}.$$

Note that, as $T \rightarrow 0$,

$$U \rightarrow \varepsilon N,$$

that is, all particles are in the low energy state, $U_1 = \varepsilon$.

As, $T \rightarrow \infty$,

$$U \rightarrow \frac{3}{2} \varepsilon N,$$

that is, half the particles are in the low energy state, $U_1 = \varepsilon$, and half the particles are in the high energy state, $U_2 = 2\varepsilon$.

Remark: We make two observations from the last example. First, the statistical temperature is always positive. Second, for each N and T , the temperature relation can be inverted for U , i.e., there is a function $\tilde{U} = \tilde{U}(T, N)$, such that

$$\tilde{U}(T(U, N), N) = U, \quad \forall (U, N) \in \text{dom}(T)$$

Thus, there is a function

$$\tilde{S}_{\text{eq}} = \tilde{S}_{\text{eq}}(T, N)$$

such that

$$S_{\text{eq}}(U, N) = \tilde{S}_{\text{eq}}(T(U, N), N)$$

for all $(U, N) \in \text{dom}(T)$.

We will assume this holds for the general case.

Definition (10.5): The statistical Helmholtz free energy is defined as

$$(10.11) \quad F(T, N) := \tilde{U}(T, N) - T \tilde{S}_{eq}(T, N),$$

for all $(T, N) \in \text{dom}(\tilde{U})$.

Theorem (10.6): For the present case (N non-interacting particles with internal energy U), it follows that

$$(10.12) \quad F(T, N) = -N k_B T \ln(Z),$$

where the partition function satisfies

$$(10.13) \quad Z = \sum_{i=1}^r \exp\left(\frac{-U_i}{k_B T}\right).$$

Proof: Recall that, from (10.1),

$$S_{eq}(U, N) = \frac{U}{T(U, N)} + k_B \log(Z) N$$

with

$$Z = \sum_{i=1}^r \exp\left(\frac{-U_i}{k_B T(U, N)}\right).$$

Then,

$$\tilde{S}_{eq}(T, N) = \frac{\tilde{U}(T, N)}{T} + k_B \log(Z) N$$

where the partition function is given by (10.13).

Using the definition (10.11), the result follows. |||

Theorem (10.7): For the present case (N non-interacting particles with internal energy U), it follows that

$$(10.14) \quad \tilde{U}(T, N) = N k_B T^2 \frac{\partial}{\partial T} [\log(Z)]$$

and

$$(10.15) \quad \tilde{S}_{eq}(T, N) = N k_B \log(Z) + N k_B T \frac{\partial}{\partial T} [\log(Z)],$$

where

$$(10.16) \quad Z = \sum_{i=1}^r \exp\left(\frac{-U_i}{k_B T}\right).$$

Proof: Differentiating $\log(Z)$ with respect to T , we have

$$\begin{aligned} \frac{\partial}{\partial T} [\log(Z)] &= \frac{1}{Z} \frac{\partial Z}{\partial T} \\ &= \frac{1}{Z} \sum_{i=1}^r \exp\left(\frac{-U_i}{k_B T}\right) \frac{U_i}{k_B T^2}. \end{aligned}$$

Thus,

$$N k_B T^2 \frac{\partial}{\partial T} [\log(Z)] = \frac{N}{Z} \sum_{i=1}^r \exp\left(\frac{-U_i}{k_B T}\right) U_i$$

$$(9.9) \quad = \sum_{i=1}^r n_i^{eq} U_i$$

$$(9.8) \quad = U.$$

Thus

$$\tilde{U}(T, N) = N k_B \frac{\partial}{\partial T} [\log(Z)].$$

Finally, because

$$F(T, N) := \tilde{U}(T, N) - T \tilde{S}_{eq}(T, N)$$

and

$$F(T, N) = -N k_B T \ln(Z),$$

as we observed in the previous result, it follows that

$$\tilde{S}_{eq}(T, N) = N k_B \log(Z) + N k_B T \frac{\partial}{\partial T} [\log(Z)],$$

using (10.14) and (10.12). ///